



COMPETENCY STANDARD
FOR
Multitasking Operation for Footwear Manufacturing
Leather and Footwear Industry Sector

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

Copyright

The Competency Standards for Multitasking Operation for Footwear is a document for developing curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the industry's requirements for individuals who pass through the set standard via assessment. Subsequently, they would be qualified and settled for a relevant job.

The document has been developed to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)** to meet the skills requirements of the Leather and Footwear industry sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

Public and private institutes may use the information contained in this competency standard for training activities benefitting Bangladesh.

Other interested parties must obtain permission from the owner of this document for reproduction of information in any manner in whole or in part of this Skills Standard, in English or other languages.

This document is available at:

Skills for Industry Competitiveness and Innovation Program (SICIP)

Finance Division, Ministry of Finance,

Probashi Kalyan Bhaban (Level 15 & 16)

71-72 Eskaton Garden

Dhaka-1000

Phone: +8802 55138753-55, Fax: +88 02 55138752

Website: <https://sicip.gov.bd/>

INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of Occupation-specific Experts from the leather industry, Consultants, and Program officers(QA) of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units and Elements Table

Generic – Compulsory (3 units of competency required)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LEA-MOF-01-G	Use basic mathematical methods	1. Identify calculation requirements in the workplace. 2. Select appropriate mathematical methods for the calculation. 3. Use tools and instruments to perform calculations.	8
SICIP-LEA-MOF-02-G	Carry out workplace interaction	1. Interpret workplace communication and etiquette. 2. Read and understand workplace documents. 3. Participate in workplace meetings and discussions. 4. Practice professional ethics at work.	8
SICIP-LEA-MOF-03-G	Operate in a team environment	1. Identify team goals and work processes. 2. Identify own role and responsibilities within team. 3. Communicate and co-operate with team members. 4. Practice problem solving within the team.	8
Sub-total			24

Sector-specific – Compulsory (2 units of competency required)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LEA-MOF-01-S	Apply occupational health and safety (OHS) practice in the workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety Practices 3. Report hazards and risks 4. Respond to emergencies	12
SICIP-LEA-MOF-02-S	Apply Green Practices in Leather & Footwear Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	04
Sub-total			16

Occupation-specific – Compulsory (5 units of competency required)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LEA-MOF-01-O	Perform cutting machine operations	1. Identify different types of cutting machines. 2. Identify tools, equipment and machinery. 3. Identify types of materials. 4. Prepare materials for cutting 5. Operate cutting machines.	60
SICIP-LEA-MOF-02-O	Perform setting and preparation in Footwear manufacturing	1. Use tools and equipment in setting and preparation 2. Identify materials for setting and preparation in footwear manufacturing 3. Perform different types of setting and preparation in footwear manufacturing 4. Check the setting and preparation components	20
SICIP-LEA-MOF-03-O	Perform sewing machine operations	1. Identify sewing machines and their major parts 2. Operate flatbed plain sewing machine 3. Operate post-bed plain sewing	130

Code	Unit of Competency	Elements of Competency	Duration (hours)
		machine 4. Operate flat-bed zigzag sewing machine	
SICIP-LEA-MOF-04-O	Perform lasting and finishing operations	1. Identify lasting and finishing machines. 2. Perform major lasting operations. 3. Perform finishing operations.	110
Sub-total			320
Total Hours			360

COMPETENCY STANDARD: MULTITASKING OPERATION FOR FOOTWEAR MANUFACTURING

Multitasking Operation for Footwear Manufacturing (Leather and Footwear Sector)

A. The Generic (Basic) Competencies

Unit of Competency: USE BASIC MATHEMATICAL METHODS	Nominal Duration: 8 hours	Unit Code: SICIP-LEA-CUT-01-G
Unit Descriptor: This unit of competency requires knowledge, skills and attitude required to use basic mathematical methods. It specifically includes the tasks of identifying calculation requirements in the workplace, selecting appropriate mathematical methods for the calculation and using tools and instruments to perform calculations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify calculation requirements in the workplace.	1.1. <u>Calculation requirements</u> are identified from <u>workplace information</u> . 1.2. Mathematical problems are constructed from workplace information.
2. Select appropriate mathematical methods for the calculation.	1.3. <u>Appropriate method</u> is selected to carry-out calculation requirements. 1.4. Constructed mathematical problems are solved with appropriate method.
3. Use tools and instruments to perform calculations.	1.5. <u>Tools and instruments</u> required for computation are identified. 1.6. Calculation is performed using appropriate tools and instruments accurately.

Range of variables:

Variable	Range (May include but not limited to):
1. Calculations requirements	1.1 Unit 1.2 Area 1.3 Height/length/breadth/thickness 1.4 Diameter 1.5 Capacity 1.6 Time 1.7 Temperature 1.8 Material/data usage 1.9 Speed
2. Workplace information	2.1 Floor environment 2.2 Swatch Book 2.3 Show Board 2.4 Specification sheet 2.5 Working chart/drawing 2.6 Standard operating procedure (SOP) 2.7 Job sheet
3. Appropriate method	3.1 Addition 3.2 Subtraction 3.3 Division 3.4 Multiplication 3.5 Conversion 3.6 Percentage and ratio calculation 3.7 Simple equation
4. Tools and instruments	4.1 Calculator 4.2 Cell phone

	4.3 Computer 4.4 Ruler
--	---------------------------

Curricular Content Guide

1. Underpinning Knowledge	1.1 Numerical concepts 1.2 Basic mathematical methods such as addition, subtraction, multiplication and division and percentage 1.3 Mathematical language, symbols and terminology 1.4 Measuring units
2. Underpinning Skills	2.1 Construct simple problems with workplace information 2.2 Solve problems using appropriate method and instruments 2.3 Use appropriate tools and instruments.
3. Underpinning Attitudes	3.1 Prompt in carrying out activities 3.2 Tidy and punctual 3.3 Respectful of peers, subordinates and seniors in the workplace 3.4 Safely use tools and equipment 3.5 Sincere and honest concerning duties
4. Resource Implications	The following resources must be provided. 4.1 Work place (simulated or actual) 4.2 Calculator 4.3 Cell phone 4.4 Computer/laptop/notebook 4.5 Measuring tape 4.6 Ruler 4.7 Projector 4.8 Stationary 4.9 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified calculation requirements from workplace information 1.2 Selected appropriate method to carry out the calculation requirements 1.3 Completed calculations using appropriate tools/instruments
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Oral test 2.3 Observation 2.4 Demonstration 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be done in a training institute or an actual or simulated workplace after completion of this unit of competency. 3.2 Assessment must be done by a suitably qualified/certified assessor.

Unit of Competency: CARRY OUT WORKPLACE INTERACTION	Nominal Duration: 8 hrs.	Unit Code: SICIP-LEA-CUT-02-G
Unit Descriptor:		

This unit of competency requires knowledge, skills and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a <u>courteous manner</u> to gather and convey information. 1.4 <u>Workplace procedures and matters</u> are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from <u>appropriate sources</u> . 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	2.5 Team meetings are attended on time. 2.6 Meeting procedures and etiquette are followed. 2.7 Active participation is ensured, opinions are expressed and heard. 2.8 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	2.9 Responsibilities as a team member are performed. 2.10 Tasks are performed in accordance with workplace procedures. 2.11 Confidentiality is maintained. 2.12 Inappropriate and conflicting situations are avoided.

Range of Variables

Variable	Range (May include but not limited to):
1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	2.1 Notes 2.2 Arranging a meeting 2.3 Agenda 2.4 Simple reports such as progress and incident reports 2.5 Job sheets 2.6 Operational manuals 2.7 Brochures and promotional material 2.8 Visual and graphic materials 2.9 Standards 2.10 OHS information 2.11 Signs
3. Appropriate sources	2.1 Human Resources (HR) Department 2.2 Managers

	2.3 Supervisors 2.4 Management Information System (MIS)
--	--

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Workplace communication and etiquette 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette 1.4 Professional ethics
2. Underpinning Skills	1.5 Demonstrating workplace communication and etiquette 1.6 Interpreting workplace instructions and symbols 1.7 Demonstrating active participation in workplace meeting 1.8 Applying professional ethics at work
3. Underpinning Attitudes	3.1 Prompt in carrying out activities 3.2 Tidy and punctual 3.3 Respectful of peers, subordinates and seniors in the workplace 3.4 Concerned about the work environment 3.5 Sincere and honest concerning duties
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Codes of conduct 4.6 Projector 4.7 Stationary 4.8 Learning manual

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Demonstration 2.3 Oral test 2.4 Observation 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be done in a training institute or an actual or simulated workplace after completion of this unit of competency. 3.2 Assessment must be done by a suitably qualified/certified assessor.

Unit of Competency: OPERATE IN A TEAM ENVIRONMENT	Nominal Duration: 8 hrs.	Unit Code: SICIP-LEA-CUT-03-G
Unit Descriptor: This unit of competency requires knowledge, skills and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, Identifying own role and responsibilities within team, communicating and cooperating with team members and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Roles and objectives of the team are identified and interpreted. 1.2 Roles and responsibilities of team members are identified and interpreted.
2. Identify own role and responsibilities within team	2.1 Personal role and responsibilities are identified within the team environment. 2.2 Reporting relationships are interpreted within team and external to team.
3. Communicate and cooperate with team members	3.1 Other teammates' tasks are identified and support provided when requested. 3.2 The team is encouraged through <u>sharing information</u> or expertise, working together to solve problems, and putting team success first. 3.3 Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1 Problems faced at the individual and team level are identified and showed insight into the root causes of the problems. 4.2 A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3 The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4 It is looked beyond the obvious and not stopped at the first answers.

Range of Variables

Variable	Range (May include but not limited to):
1.Sharing information	1.1 Agenda 1.2 Minutes 1.3 Progress and incident reports 1.4 Operational manuals 1.5 Visual and graphic materials 1.6 Emails and SMS 1.7 Phone directory 1.8 Policy, procedure and standards 1.9 OHS information

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and work processes 1.2 Roles and responsibilities 1.3 Finding problems and solving them
2. Underpinning Skills	2.1 Identifying own role and responsibilities within team 2.2 Communicating and cooperating with team members 2.3 Demonstrating problem solving within the team
3. Underpinning Attitudes	2.1 Active on teamwork 2.2 Prompt in carrying out activities 2.3 Tidy and punctual 2.4 Respectful of peers, subordinates and seniors in the workplace 2.5 Sincere and honest concerning duties
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationary 4.4 Learning manual

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified team goals and work processes 1.2 Communicated and cooperated with team members. 1.3 Worked as a team member 1.4 Solved problems as a team member
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Demonstration 2.3 Oral test 2.4 Observation 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be done in a training institute or an actual or simulated workplace after completion of this unit of competency. 3.2 Assessment must be done by a suitably qualified/certified assessor.

B. The Sector Specific (Common) Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 12 hrs.	Unit Code: SICIP-LEA-MOF-01-S
Unit Descriptor: This unit of competency requires knowledge, skills and attitudes required to apply occupational health and safety (OHS) practice in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted. 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	1.4 OHS policies and procedures are applied in the workplace including <u>personal protective equipment (PPE)</u> . 1.5 Common health issues are recognized. 1.6 Common safety issues are identified.
3. Report hazards and risks	1.7 <u>Hazards and risks</u> are identified. 1.8 Hazards and risks assessment and controls are interpreted.
4. Respond to emergencies	1.9 Alarms Respond to alarms and warning devices. 1.10 <u>Emergency response plans and procedures</u> are responded to. 1.11 <u>First aid procedures</u> during emergency situations are identified.

Range of Variables

Variable	Range (May include but not limited to):
1. OHS policies	1.1 Organizational OHS polices 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal Protective Equipment (PPE)	1.4 Safety glasses 1.5 Ear plugs 1.6 Gloves 1.7 Apron 1.8 Mask
3. Hazards and risks	1.1 Chemical hazards. 1.2 Biological hazards. 1.3 Physical Hazards. 1.3.1 Machine hazards. 1.3.2 Materials hazards. 1.3.3 Tools and Equipment hazards.
4. Emergency response plans and procedures	1.4 Firefighting procedures 1.5 Earthquake response procedures 1.6 Emergency response plans and procedures 1.7 Medical and first aid
5. First aid procedure	1.8 Washing of open wound

	1.9 Washing chemically infected area 1.10 Applying bandage 1.11 Taking appropriate medicine
--	---

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE)
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Applying personal health and safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergencies
3. Underpinning Attitudes	3.1 Committed to occupational health and safety practices 3.2 Communicates well with peers, subordinates and seniors in workplace 3.3 Prompt in carrying out activities 3.4 Tidy and punctual 3.5 Sincere and honest concerning duties 3.6 Responsible during emergencies
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits 4.6 Projector 4.7 Stationary 4.8 Learning manual

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies
3. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Demonstration 2.3 Oral test 2.4 Observation 2.5 Portfolio
4. Context of Assessment	3.1 Competency assessment must be done in a training institute or an actual or simulated workplace after completion of this unit of competency. 3.2 Assessment must be done by a suitably qualified/certified assessor.

Unit of Competency: APPLY GREEN PRACTICES	Nominal Duration: 04 hrs.	Unit Code: SICIP-LEA-MOF-01-S
Unit Descriptor: This unit of competency requires knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices</u> are explained. 1.2 <u>Sources of environmental impacts</u> during footwear activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 <u>Ways of mitigating environmental impacts</u> in leather sector are explained.
2. Minimize resource use in the workplace	1.1 Water, energy, and raw material consumption are documented. 1.2 <u>Recyclable and non-recyclable items</u> are identified. 1.3 Procedures to reduce resource consumption are implemented. 1.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	1.5 <u>Different types of waste</u> are identified. 1.6 Hazardous waste is disposed of according to environmental regulations. 1.7 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range (May include but not limited to):
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair 1.3 use of sustainable materials with low environmental impact 1.4 Recycling materials 1.1 Sustainable transportation
2. Sources of environmental impacts	1.5 Air pollution 1.5.1 Dust generation 1.5.2 Emission from machinery and vehicles 1.6 Water pollution 1.6.1 Debris and sediments 1.6.2 Chemical's leakage 1.7 Noise pollution 1.8 Waste generation 1.8.1 waste

	1.8.2 Non-recyclable and hazardous materials 1.9 Resource Depletion 1.9.1 Excessive use of raw materials 1.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	1.10 Utilizing Energy-Efficient Equipment 1.11 Adopting Renewable Energy Sources 1.12 Implementing Site Protection Measures 1.13 Using Reusable Materials 1.14 Choosing Sustainable Materials 1.15 Using Noise-Reducing Equipment and scheduling work appropriately 1.3 Optimizing Logistics and Delivery
4. Recyclable and non-recyclable items	1.16 Recyclable items 1.16.1 Metal 1.16.2 Wood 1.16.3 Brick 1.16.4 Concrete 1.17 Non-recyclable items 1.17.1 Paints and coatings 1.17.2 Contaminated materials like lead paint, asbestos 1.4 Treated wood
5. Different types of waste	1.18 Wood waste 1.19 Metal waste 1.20 Plastic waste 1.21 Electrical system waste 1.22 Gypsum board waste 1.23 Packaging waste 1.24 Organic waste 1.25 Chemical West 1.5 Fuel/Oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in Leather sector 1.2 Key terms and symbols related to environmental sustainability in Ship drawings 1.3 Sources of environmental impacts in Shipbuilding 1.4 Methods to minimize resource consumption (water, energy, raw materials) 1.5 Waste management practices 1.6 Sustainable alternatives to fossil-based energy resources 1.7 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	1.1 Identifying environmental impacts in Leather activities 1.2 Applying methods to minimize resource use in the workplace 1.3 Implementing waste management practices 1.4 Following procedures to safely dispose of hazardous materials 1.5 Documenting water, energy, and material consumption in the workplace 1.6 Performing green practices in personal and professional activities
3. Underpinning Attitudes	3.1 Commitment to sustainable and eco-friendly practices 3.2 Active participation in reducing resource use and waste generation

	3.3 Eagerness to adopt green habits in professional life 3.4 Promptness in implementing sustainable practices in activities 3.5 Tidiness and concern for environmental cleanliness 3.6 Compliance with safety standards and environmental regulations
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of hand tools and power tools 4.3 Safety gear 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Operation and maintenance manuals 4.8 Waste segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Explained the principles of green practices in Shipbuilding 1.2 Identified sources of environmental impacts 1.3 Implemented procedures to minimize resource use (water, energy, raw materials) 1.4 Disposed of hazardous waste in compliance with safety and environmental standards 1.5 Practiced green habits to reduce waste in both personal and professional life
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

C. Occupation-specific Competencies

Unit of Competency: PERFORM CUTTING MACHINE OPERATIONS	Nominal Duration: 60 hrs.	Unit Code: SICIP-LEA-CUT-01-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitudes required to perform cutting machine operations. It specifically includes the tasks of identifying different types of cutting machines, identifying tools, equipment and machinery, identifying types of materials, preparing materials for cutting and operating cutting machines.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Identify different types of cutting machines.	1.1 <u>Types of cutting machines</u> are identified 1.2 <u>Principles</u> of cutting machine operation are explained. 1.3 <u>Types of cutting</u> are identified and described. 1.4 Quality of cutters and clickers are identified.
2. Identify tools, equipment and machinery	2.1 Common <u>tools and equipment</u> used are identified. 2.2 Common machinery used is identified.
3. Identify types of materials.	3.1 <u>Types of materials</u> used in cutting are identified 3.2 <u>Associated materials</u> used in cutting are identified
4. Prepare materials for cutting	4.1 Layer making is prepared 4.2 Prepared layers are preserved for 24 hours 4.3 <u>Defect of natural (Leather) materials</u> are identified 4.4 Materials are sorted out according to size, grain pattern, shade, thickness and finish.
5. Operate cutting machines.	5.1 Cutting Knife is selected as per the job specification 5.2 Cutting machine is set up as per SOP 5.3 Interlocking/Nesting is carried out as per SOP 5.4 Cutting is carried out as per job sheet 5.5 Defective Pieces are identified and marked

Range of Variables

Variable	Range (Includes but not limited to):
1. Types of cutting machines	1.1 Beam cutting machine 1.2 Swing arm clicking press 1.3 Travelling head clicking press (THCP)
2. Principles	2.1 Set-up machine according to knife height 2.2 Ensure proper and sufficient lighting 2.3 Ensure cutting knife is at required sharpness 2.4 Examine both sides of skin to identify defects 2.5 Consider quality requirements of shoe parts 2.6 Level the cutting board as per job requirements 2.7 Ensure no overcut and undercut
3. Types of materials	3.1 Leather 3.2 Synthetics 3.3 Fabrics 3.4 Reinforcements/interlinings 3.5 Sheet
4. Associated materials	4.1 Upper leather 4.2 Lining

	4.3 Sole 4.4 Insole board 4.5 Synthetics 4.6 Fabrics 4.7 Foams 4.8 Reinforcements
5. Defect of natural (Leather) materials	5.1 Natural: 5.1.1 Loose grain 5.1.2 Growth marks 5.1.3 Pinhole 5.1.4 Tick marks 5.1.5 Skin disease 5.1.6 Vein marks 5.1.7 Grain damage 5.1.8 Pox marks 5.1.9 Scratch marks 5.2 Man-made: 5.2.1 Flay cuts 5.2.2 Salt stain 5.2.3 Brand marks 5.2.4 Color/shade variation 5.2.5 Grain pattern variation 5.2.6 Wound mark 5.2.7 Looseness 5.3 Grain jam
6. Types of cutting	6.1 Hand 6.2 Machine
7. Tools and equipment	1.1 Cutting dies/knives 1.2 Scissors 1.3 Thickness gauge 1.4 Hammer 1.5 Marker 1.6 Cutting was

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of cutting machine operations 1.2 Types of materials 1.3 Tools, equipment and machinery 1.4 Trims and accessories
2. Underpinning Skills	2.1 Explaining principles of cutting machine operations 2.2 Interlocking/ nesting of workpiece 2.3 Operating cutting machines
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Concern to environmental care 3.3 Eagerness to learn 3.4 Tidiness, timeliness, and orderliness 3.5 Respect for rights of peers and seniors in workplace 3.6 Communication with peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Tools and equipment 4.3 Cutting machine 4.4 Material

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Explained principles of cutting machine operations 1.2 Identified tools, equipment and machinery for cutting 1.3 Performed interlocking/ nesting 1.4 Operated cutting machines
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Oral test 2.3 Observation 2.4 Demonstration 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be done in a training institute or an actual or simulated workplace after completion of this unit of competency. 3.2 Assessment must be done by a suitably qualified/certified assessor.

Unit of Competency: PERFORM SETTING AND PREPARATION IN FOOTWEAR MANUFACTURING	Nominal Duration: 20 hrs.	Unit Code: SICIP-LEA-MOF-02-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitudes required to perform setting and preparation in footwear manufacturing. It specifically includes the tasks of using tools and equipment in setting and preparation, identifying materials for setting and preparation in footwear manufacturing and performing different types of setting and preparation in footwear manufacturing.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Use tools and equipment in setting and preparation	1.1 <u>Tools</u> for setting operations are identified 1.2 Applications of tools and equipment is recognized 1.3 Usability of tools and equipment is checked and verified 1.4 Unsafe of faulty tools and equipment is identified and marked for repair
2. Identify materials for setting and preparation in footwear manufacturing	2.1 <u>Materials</u> for setting operations are identified 2.2 Applications of materials is recognized 2.3 Usability of materials is checked and verified 2.4 Unsafe materials are identified and marked for repair.
3. Perform different types of setting and preparation in footwear manufacturing	2.5 Table, tools, machines and work area are prepared as per the workplace standards 2.6 <u>Setting and preparation works</u> are performed according to job specification 2.7 The components of setting and preparation are checked and verified

Range of Variables

Variable	Range (Includes but not limited to):
1. Tools	1.1 Setting Gauge 1.2 Folding pattern 1.3 Skiving Gauge 1.4 Measuring Scale 1.5 Thickness Gauge 1.6 Folding Guide 1.7 Setting Pattern 1.8 Awls 1.9 Hammer 1.10 Punches
2. Materials	2.1 Upper and Lining 2.2 Folding Cord 2.3 Reinforcement Tape 2.4 Reinforcement Fabric Materials 2.5 Adhesive
3. Setting and preparation works	3.1 Fusing 3.2 Marking 3.3 Skiving 3.4 Folding 3.5 Adhesive application 3.6 Embossing

Curricular Content Guide

1. Underpinning Knowledge	1.1 Different types of materials 1.2 Setting and preparation works
2. Underpinning Skills	2.1 Preparing materials 2.2 Setting and preparation of footwear 2.3 Checking quality
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Concern to environmental care 3.3 Eagerness to learn 3.4 Tidiness, timeliness, and orderliness 3.5 Respect for rights of peers and seniors in workplace 3.6 Communication with peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Materials 4.5 Job specification

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Workplace (simulated or actual) 1.2 Personal protective equipment (PPE) 1.3 Tools and equipment 1.4 Materials 1.5 Job specification
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Oral test 2.3 Observation 2.4 Demonstration 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be done in a training institute or an actual or simulated workplace after completion of this unit of competency. 3.2 Assessment must be done by a suitably qualified/certified assessor.

Unit of Competency: PERFORM SEWING MACHINE OPERATIONS	Nominal Duration: 130 hrs	Unit Code: SICIP-LEA-CUT-03-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitudes required to perform sewing machine operations. It specifically includes the tasks of identifying sewing machines and their major parts, operating flatbed plain sewing machine, operating post-bed plain sewing machine and operating flat-bed zigzag sewing machine.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Identify sewing machines and their major parts	1.1 <u>Types of sewing machines</u> are identified 1.2 The <u>major parts of sewing machine</u> are identified 1.3 <u>Needles and threads</u> are identified
2. Operate flatbed plain sewing machine	2.1 Job specification and instructions are read and interpreted 2.2 Appropriate PPE, tools and equipment is identified and selected 2.3 Thread tension, stitch length and allowance guide are adjusted as per SOP 2.4 Flatbed sewing machine is operated as per the job specifications 2.5 Sewing components are checked and verified
3. Operate post-bed plain sewing machine	3.1 Job specification and instructions are read and interpreted 3.2 Appropriate PPE, tools and equipment is identified and selected 3.3 Thread tension, stitch length and allowance guide are adjusted as per SOP 3.4 Post bed sewing machine is operated as per the job specifications 3.5 Sewing components are checked and verified
4. Operate flat-bed zigzag sewing machine	4.1 Job specification and instructions are read and interpreted 4.2 Appropriate PPE, tools and equipment is identified and selected 4.3 Thread tension, stitch length and allowance guide are adjusted as per SOP 4.4 Flatbed zigzag sewing machine is operated as per the job specifications 4.5 Sewing components are checked and verified

Range of Variables

Variable	Range (Includes but not limited to):
1. Types of sewing machines	1.1 Flatbed single needle sewing machine 1.2 Flatbed double needle sewing machine 1.3 Post bed single needle sewing machine 1.4 Post bed double needle sewing machine 1.5 Flatbed zigzag sewing machine 1.6 Cylinder bed zigzag sewing machine 1.7 Cylinder bed plain binding machine 1.8 Stroble sewing machine
2. Major Parts	2.1 Spool pin 2.2 Thread guide

	2.3 Tension disc 2.4 Take-up lever 2.5 Needle bar 2.6 Bobbin case 2.7 Presser foot/pressure wheel 2.8 Presser foot lifter 2.9 Stitch regulator 2.10 Bobbin winder 2.11 Fly wheel/balance wheel 2.12 Clutch 2.13 Slide plate 2.14 Needle plate/throat plate 2.15 Feed dog 2.16 Face plate 2.17 Spool pin (for bobbin winding)
3. Needles and threads	Needle 2.1 Ball Point 2.2 Cut Point 2.3 Needle Number Thread 2.4 Polyester 2.5 Nylon 2.6 Cotton 2.7 Thread Number

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of sewing machines 1.2 Major parts of sewing machines 1.3 Needles and threads 1.4 Operating principles of flatbed plain sewing machine 1.5 Operating principles of post-bed plain sewing machine 1.6 Operating principles of flat-bed zigzag sewing machine
2. Underpinning Skills	2.1 Operating flatbed plain sewing machine 2.2 Operating post-bed plain sewing machine 2.3 Operating flat-bed zigzag sewing machine
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Concern to environmental care 3.3 Eagerness to learn 3.4 Tidiness, timeliness, and orderliness 3.5 Respect for rights of peers and seniors in workplace 3.6 Communication with peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Sewing machines 4.4 Leather samples (various)

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Operated flatbed plain sewing machine 1.2 Operated post-bed plain sewing machine 1.3 Operated flat-bed zigzag sewing machine
2. Methods of Assessment	Competency should be assessed by:

	2.1 Written test 2.2 Oral test 2.3 Observation 2.4 Demonstration 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be done in a training institute or an actual or simulated workplace after completion of this unit of competency. 3.2 Assessment must be done by a suitably qualified/certified assessor.

Unit of Competency: PERFORM LASTING AND FINISHING OPERATIONS	Nominal Duration: 110 hrs	Unit Code: SICIP-LEA-CUT-04-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitudes required to perform lasting and finishing operations. It specifically includes the tasks of identifying lasting and finishing machine, performing major lasting operations and performing finishing operations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Identify lasting and finishing Machine	1.1 <u>Types of lasting machines</u> are identified 1.2 Appropriate personal protective equipment (PPE) is identified and selected. 1.3 Appropriate <u>tools and equipment</u> are identified and selected.
2. Perform major lasting operations	2.1 Appropriate PPE, tools and equipment is identified and selected 2.2 Appropriate molds are selected and pressure, time and temperature are adjusted as per SOP 2.3 Appropriate pincher, toe band and wafer plate are selected, and pressure, time and temperature are adjusted as per SOP 2.4 Appropriate heel band and wafer plate are selected, and pressure, time and temperature are adjusted as per SOP 2.5 Suitable grade of emery paper is selected, and RPM is adjusted as per the jobs 2.6 <u>Adhesives</u> and <u>primer</u> are used appropriately, and temperature of heat reactivator and speed are maintained as per the jobs 2.7 Negative molds are selected appropriately, and timing and pressure are set as per the jobs
3. Perform Finishing Operations	3.1 Number of suitable nails are selected and pressure is adjusted 3.2 Suitable brushes, wax and cream are selected and machine RPM is set as per SOP 3.3 Finishing operations are carried out as per the job requirements 3.4 Finished shoes are checked and verified as per the jobs

Range of Variables

Variable	Range (Includes but not limited to):
1. Types of lasting machines	1.1 Back part molding 1.2 Toe puff steaming machine 1.3 Toe lasting machine 1.4 Seat and side lasting machine 1.5 Heel lasting machine 1.6 Marking machine 1.7 Pounding Machine 1.8 Roughing machine 1.9 Hot air Machine 1.10 Sole press machine 1.11 Nailing Machine 1.12 Brushing machine

Variable	Range (Includes but not limited to):
2 Tools and equipment	2.1 Back part counter molding mold 2.2 Pincher and wafer plate 2.3 Hand Pincher 2.4 Hand Roughing 2.5 Pneumatic hammer 2.6 Roughing Pulley 2.7 Sole pressing negative mold 2.8 Polishing Brush 2.9 Manual Hammer
3 Adhesives	3.1 Neoprene 3.2 Poly Urethane(PU) 3.3 Latex 3.4 Hot Melt
4 Primer	4.1 PU Primer 4.2 PVC Primer 4.3 Rubber Primer 4.4 TPR Primer 4.5 Leather Primer 4.6 EVA Primer

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of lasting machines 1.2 Tools and equipment 1.3 Adhesive and primers 1.4 Process of lasting operation 1.5 Process of finishing operation
2. Underpinning Skills	2.1 Selecting appropriate mold 2.2 Selecting pinchers, toe band, and wafer plate 2.3 Adjusting pressure, time and temperature in machines 2.4 Selecting appropriate heel bands 2.5 Selecting appropriate emery paper 2.6 Using heat reactivator to bond adhesives 2.7 Installing fasteners, rivets and other attachments 2.8 Conducting surface treatments 2.9 Adding embossing or debossing, branding and decorative elements 2.10 Conducting final inspection
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Concern to environmental care 3.3 Eagerness to learn 3.4 Tidiness, timeliness, and orderliness 3.5 Respect for rights of peers and seniors in workplace 3.6 Communication with peers and seniors in workplace
4. Resource Implications	1.1 Workplace (simulated or actual) 1.2 Personal protective equipment (PPE) 1.3 Tools and equipment

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Performed major lasting operations 1.2 Perform finishing operations
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Oral test

	2.3 Observation 2.4 Demonstration 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be done in a training institute or an actual or simulated workplace after completion of this unit of competency. 3.2 Assessment must be done by a suitably qualified/certified assessor.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Multitasking Operation for Footwear Manufacturing
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom and Workshop/ Lab):

- Classroom : 350 sq. ft.
- Workshop/Lab : 800-1000 sq. ft.

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Beam cutting machine	1
2.	Swing arm clicking press	1
3.	Travelling head clicking press (THCP)	1
4.	Skiving Machine	2
5.	Folding Machine	1
6.	Fusing Machine	1
7.	Emboss Machine	1
8.	Flatbed Sewing Machine	2
9.	Post bed Sewing machine	2
10.	Flatbed Zigzag Sewing Machine	1
11.	Counter Molding Machine	1
12.	Toe Lasting Machine	1
13.	Seat and Side/ Heel Lasting Machine	1
14.	Roughing Machine	1
15.	Heat Reactivator	1
16.	Sole press machine	1
17.	Nailing Machine	1
18.	Polishing Machine	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For running the training course on Multitasking Operation, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.